VERIFIED PROGRESS REPORT

ATLAS Integration Analysis

Concrete work completed during the hardware integration period, plus what remains unfinished.

> Evidence language
> Confirmed means supported by code, a saved setup record, or a completed test. Likely means a strong inference. Assumption identifies something that still needs a physical check.

Prepared 3 August 2026

Project ATLAS School Pilot v1 - Team Touchdown

1. Answer

> Current verdict
> Confirmed: ATLAS is now a working integrated prototype for the Jetson Orin NX, wireless helmet camera, Shokz headset, YOLO/TensorRT vision, grounded RAG, Gemini dialogue, cloud speech, offline fallbacks, and a local dashboard. It is not a finished exhibition system: the EV3 artwork stands, camera battery assembly, Shokz button capture, and several deployment hardening tasks remain incomplete.

Why this conclusion

- Confirmed: Camera, Shokz audio, CUDA, YOLO, RAG, Gemini, speech providers, and dashboard paths were exercised on the Jetson.
- Confirmed: Two complete camera-to-spoken-answer cycles were completed before the cloud-speech upgrade: one mock-LLM cycle and one real Gemini cycle.
- Confirmed: The latest complete Jetson regression run before the live dashboard camera addition passed 174 tests; the camera-panel change then passed 40 focused tests and lint.
- Confirmed: Physical motor control remains disabled in configuration and was not tested on the new Jetson.

2. Reporting scope and evidence

This report covers the verified integration sequence recorded on 1-2 August 2026 and the final state carried into 3 August. It intentionally excludes ideas that were only discussed. Sources are listed in section 8. The Jetson was powered off before this report was written, so no new live hardware claims were added.

Evidence class	What counts	How it is used
Confirmed	Saved logs, current source code/configuration, completed Jetson command output, or passing tests.	Reported as completed or implemented.
Likely	Strong inference from implementation, but no final physical test.	Reported with a clear limitation.
Assumption	Planned behavior or hardware state that still needs measurement.	Placed only in the unfinished-work section.

3. What was concretely completed

Jetson Orin NX platform and recovery

- Confirmed: The Seeed reComputer Super J401 / Orin NX 16 GB booted and was usable after a partial NVIDIA L4T upgrade failure.
- Confirmed: A no-reflash dpkg recovery was applied; apt checks and dpkg audit were clean after reboot.
- Confirmed: Critical NVIDIA L4T packages were held to prevent another generic upgrade from repeating the board mismatch failure.
- Confirmed: CUDA-enabled PyTorch 2.8.0 loaded on the Orin GPU; compatible NumPy, OpenCV, SciPy, Ultralytics, audio, RAG, and Gemini packages were installed.

> Important platform debt
> Confirmed: The system reports Seeed L4T 36.4.3 while running a later kernel package after the interrupted upgrade. It boots and passed package checks, but this is a recovery state, not a clean vendor image.

Shokz OpenComm2 UC audio

- Confirmed: The Loop120 USB dongle was detected for microphone input and headset output.
- Confirmed: Live microphone-to-headset loopback was reported as nearly immediate and clean.
- Confirmed: Playback routing was changed to prefer the named PulseAudio Shokz sink, with ALSA fallback; this fixed a device-busy failure.
- Confirmed: Capture selection ignores PulseAudio monitor sources and chooses the actual mono microphone.
- Confirmed: A language-neutral two-note listening cue now plays immediately before recording.

XIAO ESP32-S3 Sense camera

- Confirmed: Camera firmware was configured for a 2.4 GHz network, built, flashed, and served MJPEG over Wi-Fi.
- Confirmed: The camera announces `atlas-camera.local` through mDNS and streams at `:81/stream`.
- Confirmed: Saved measurements show 23.62 FPS raw and 22.75 FPS with desktop rendering at 640x480.
- Confirmed: A fresh frame still arrived after physical handling; the camera assembly and antenna were visually checked.

Vision and TensorRT

- Confirmed: The trained detector recognizes Mona Lisa, The Starry Night, and Tutankhamun's mask.
- Confirmed: Live trials recognized Mona Lisa and the mask; the mask was observed around 50-90% confidence.
- Confirmed: TensorRT FP16 is preferred when the engine exists, with automatic PyTorch fallback.
- Confirmed: Recorded benchmark: 14.31 ms TensorRT median versus 37.52 ms PyTorch median, a measured 2.62x reduction in median wall time.
- Confirmed: A three-artwork parity check returned the same correct result with both backends.

End-to-end interaction

Camera fresh frame	->	YOLO center + hold	->	Cue visitor signal	->	STT question	->	RAG + LLM grounded answer	->	TTS Shokz output

- Confirmed: A complete mock-LLM cycle ran through camera, centered Mona Lisa hold, cue, Shokz microphone, Whisper, RAG, Piper, and clean exit.
- Confirmed: A complete real-Gemini cycle ran through the same hardware path; only transcript text and retrieved facts were sent to Gemini.
- Confirmed: Mona Lisa was detected at 86-97%, and the configured two-second centered hold triggered in about 2.0-2.1 seconds.

Cloud speech and low-latency dialogue

- Confirmed: Deepgram Nova-3 streaming STT, Cartesia Sonic 3.5 streaming TTS, and local Silero VAD are implemented and selected in the current configuration.
- Confirmed: Live Cartesia tests produced first audio in roughly 109-150 ms and played through Shokz.
- Confirmed: Local Whisper and Piper are preloaded as fallbacks; normal boot is configured to use cached models without Hub checks.
- Confirmed: Gemini token streaming is split into complete, validated sentences so TTS can speak one sentence while the next is generated.
- Confirmed: Gemini thinking was disabled for the low-latency path; a typed French Mona Lisa correction test returned a grounded answer in 913 ms.
- Confirmed: A cautious French homophone repair handles the observed qui appelle la Joconde transcription as likely qui a peint la Joconde.

RAG and content

- Confirmed: The content pack now contains seven artworks and 143 short source-attributed chunks.
- Confirmed: Four works were added: Sunflowers, Liberty Leading the People, Girl with a Pearl Earring, and The Great Wave off Kanagawa.
- Confirmed: Hybrid retrieval combines dense search, SQLite FTS5/BM25, reciprocal-rank fusion, metadata filtering, reranking, and bounded context packing.
- Confirmed: The saved detailed RAG evaluation passed 9/9 checks.

Dashboard and operational control

- Confirmed: Visitor and admin dashboards are implemented as a local FastAPI application on port 8765.
- Confirmed: The visitor page supports session controls, language/profile settings, typed questions, manual capture, artwork status, confidence, and answer latency.
- Confirmed: The admin page exposes health, privacy, configuration, RAG evaluation, logs, and emergency-stop controls; protected actions require an admin token.
- Confirmed: A live dashboard camera panel now refreshes at about eight frames per second and draws the existing YOLO box, label, and confidence without running inference twice.
- Confirmed: The final camera endpoint returned HTTP 200 with a real JPEG, and GNOME Web remained active on the Jetson display until shutdown.

Privacy and resilience

- Confirmed: Raw audio, raw images, face data, and student names are disabled; transcript logging is off by default.
- Confirmed: API keys are stored outside Git in a private environment file and are not printed by the application.
- Confirmed: Cloud STT/TTS and Gemini have local or mock fallbacks; hardware adapters fail without collapsing dev mode.

4. Verification summary

Check	Observed result	Interpretation
Complete Jetson suite	174 passed, 1 non-blocking Starlette warning	Broad regression coverage before the final camera-panel addition.
Final dashboard/vision focus	40 passed; Ruff clean	Covers tracker, dashboard API, and final camera-overlay change.
Local independent audit	173 passed; 2 unexecuted because this Windows report environment lacks cv2	No additional code regressions found; not a replacement for Jetson hardware tests.
Camera endpoint	HTTP 200; JPEG returned	Shared in-memory frame path works.
Dashboard process	GNOME Web active on DISPLAY=:1	Local on-device operator view works.
Real interaction	Mock and real-Gemini cycles completed	Core camera-to-spoken-answer workflow worked before cloud-speech replacement.

5. What has not been completed

Open item	Current evidence	Required completion check
EV3 artwork stands	Confirmed: disabled; no physical test on the Orin NX.	Pair current brick, run `ev3_motors.py`, verify A/B/C mapping and full raise/lower workflow.
Servo alternative	Confirmed: disabled and not part of the tested path.	Choose EV3 or servo architecture, then test power, movement limits, and emergency stop.
Shokz multifunction button	Confirmed: software hook exists, but no Linux button event was identified.	Capture the dongle input event and prove one press requests manual capture.
Automatic detection for four new works	Confirmed: RAG/manual capture support exists; YOLO classes do not.	Collect approved labeled images, train, export, and run parity tests.
Camera battery and heatsink	Confirmed: battery is not attached; heatsink remains off to keep BAT pads accessible.	Measure current, select protected LiPo, solder safely, then perform a 2-3 hour thermal/runtime test.
Admin token	Confirmed: not configured during the final dashboard demo.	Set `ATLAS_ADMIN_TOKEN` and test every protected action.
Remote teacher access	Confirmed: localhost and Jetson-screen use worked; another-device workflow was not proven.	Bind deliberately on a trusted network and test authentication from a teacher device.
Final full suite after camera panel	Confirmed: 40 focused tests passed; full 175-test Jetson run was not repeated after that last UI change.	Boot Jetson and run the complete suite once.
TensorRT portability	Confirmed: runtime warned that the engine plan may have been built for a different device model.	Re-export the engine directly on this Orin NX and rerun speed/parity benchmarks.
Cold boot	Confirmed: first full preload took roughly three minutes; warm restart was much faster.	Profile each preload stage and set a deployment startup target.
Long-duration reliability	Confirmed: no multi-hour soak or repeated-session endurance test is recorded.	Run camera, audio, network, and cloud reconnect tests for several hours.
Clean Jetson image	Confirmed: current system uses a no-op dpkg recovery and mixed L4T state.	Prepare a second known-good image and validate a clean reflash when schedule permits.

6. Recommended next sequence

Priority	Action	Pass condition
1	Reboot, run full suite, then run one complete Deepgram -> Gemini -> Cartesia interaction.	All tests pass and one real spoken exchange finishes cleanly.
2	Re-export TensorRT on the Orin NX.	No portability warning; parity retained; benchmark recorded.
3	Integrate EV3 stands.	Centered gaze leaves selected artwork up, lowers the others, then restores all after speech.
4	Configure admin token and test every admin operation.	Protected controls reject missing/wrong tokens and accept the correct token.
5	Complete camera battery/thermal build and soak test.	Target runtime met without brownouts, unsafe heat, or stream loss.

7. Risks

- Confirmed: The largest operational risk is the recovered mixed L4T package state; a future package operation could reopen the bootloader/kernel problem.
- Confirmed: The largest demonstration risk is unfinished physical stand control; current success is a spoken digital prototype, not the full kinetic exhibit.
- Likely: Cloud speech quality will vary with network quality and accents; local fallbacks reduce outages but do not match the same latency or recognition quality.
- Confirmed: The strongest case against the positive conclusion is that no single final run combined the latest dashboard camera overlay, cloud speech, real Gemini, and EV3. That objection survives: the prototype is integrated, but final-system acceptance is not complete.

8. Evidence register

ID	Source	Used for
E1	`ATLAS_JETSON_NX_SETUP_LOG.md`	Platform recovery, package state, installed stack, hardware checkpoints, full interactions.
E2	`docs/hardware_integration_status.md`	Camera/audio measurements, TensorRT benchmark, RAG state, incomplete hardware.
E3	`config/settings.yaml`	Current providers, privacy defaults, camera URL, disabled EV3/servo, dashboard port.
E4	`src/atlas/` current code	Streaming STT/TTS, Silero, fallback, sentence overlap, dashboard, camera overlay, manual capture.
E5	`data/content_packs/demo_pack/manifest.json`	Seven-artwork content inventory.
E6	Jetson command outputs in the integration task	174-pass complete suite, 40-pass focused suite, live endpoint and browser checks.
E7	`ATLAS_Hardware_Guide_Beginner.pdf`	Intended hardware checkpoints and comparison against unfinished work.